IV. MODELING WORKPLACE EXPOSURE

CEB has developed standard approaches for estimating exposure levels to workers via the inhalation and dermal routes and for assessing the effectiveness of controls, including ventilation, respirators, and gloves. In the absence of representative data, CEB engineers should use these approaches to quantify worker exposures.

A. Estimating Inhalation Exposures

The amount of substance inhaled by a worker is a function of many variables including the airborne concentration of the substance, the amount of time spent in an atmosphere containing the substance, the breathing rate of the worker, the worker activity or job performed, the physical and chemical properties of the substance, the temperature changes, seasonal changes and the effectiveness of engineering controls or personal protective equipment in protecting the worker. In assessing the potential for worker exposures at multiple processing and use sites, the controls used at any or all the sites may not be known. Thus, CEB engineers must estimate reasonable worst case exposures representing sites where no controls are used. If information is available about a particular site or general industry practices, the engineer also may estimate exposures assuming an effectiveness for the expected control.

This section describes the assessment of occupational exposure in the absence of respirators and engineering controls. Sections IV.C and IV.D discuss the effects of respirators and engineering controls, respectively, in reducing worker exposure.

1. General Approach

As stated previously, inhalation exposure is a function of many factors including airborne concentration, duration of exposure, inhalation rate, and effectiveness of engineering controls and personal protective equipment. Neglecting any use of controls or personal protective equipment to reduce the airborne concentration, the amount of substance available for inhalation is calculated in units of mg/day as:

$$I = C_m \; bh \qquad \text{Equation 4-1}$$

where: I = Daily inhalation exposure, mg/day

C_m = Airborne concentration of substance, mg/m^3

b = Inhalation rate, m^3/hr

h = Duration, hr/day

Determination of inhalation rate and duration of exposure is generally straightforward, and is discussed in the following paragraphs. Determination of airborne concentration can be difficult, and is discussed in detail in Section IV.A.2.

With increased physical activity, inhalation rate increases. The typical worker breathes about 10 m^3 of air in 8 hours, or 1.25 m^3/hr (NIOSH 1976). This value is slightly above the volumetric flowrate for light work (NIOSH 1976). During the workday, the volumetric flowrate at any given instant may vary widely as a function of the type of work being performed. Inhalation rates range from about 0.56 m^3/hr during rest periods to 7.9 m^3/hr during the maximum work, as shown in Table 4-1.

TABLE 4-1. INHALATION RATES

Activity	Minute volume air flow rates m^3/hr
Rest	0.56
Light work	1.18
Medium work	1.75
Med. heavy work	2.63
Heavy work	3.6
Maximum work	7.9

Source: NIOSH 1976

The duration of exposure can only be estimated on a case-by-case basis through knowledge of the activities that may lead to exposure. This information can be obtained from persons knowledgeable about the process or from descriptions of similar operations. Airborne concentrations (Permissible Exposure Levels [PELs]) or measured data are often expressed as an 8-hr Time-Weighted Average (TWA). When using an 8-hr TWA for purposes of estimating exposure, the actual duration of exposure is not used to calculate the exposure. A TWA value implies that exposure may be high for short periods of time as long as those periods are compensated by periods of lower exposure and the average exposure over 8 hours does not exceed the TWA. The maximum daily exposure is the total amount of substance to which the worker can be exposed per 8-hr shift whether the exposures are for short duration at high concentrations or long duration at low concentrations. It is calculated using a duration of 8 hr/day.

If the inhalation rate on average is 1.25 m^3/hr, the equation for calculating the amount of substance available for inhalation becomes:

$$I = 1.25h\, C_m \qquad \text{Equation 4-2}$$

When C_m is estimated as an 8-hr TWA concentration, the duration of exposure is assumed to be 8 hr/day and the equation reduces to:

$$I = 10\, C_m \qquad \text{Equation 4-3}$$

The remainder of this section discusses methods for estimating airborne concentrations for use in Equations 4-1, 4-2, or 4-3.

2. <u>Calculating Airborne Concentrations</u>

Worker inhalation exposure is best determined using personal monitoring measurements for workers performing the job under study while being exposed to "typical" pollutant levels. Section III discusses some sources of monitoring data. Examples of chemicals for which the engineering assessments have been based on industrial hygiene monitoring data include butadiene, acrylamide, and chlorinated solvents.

Since monitoring data are seldom available for PMN chemicals, CEB engineers usually rely on other methods to assess worker exposures. These methods are also used in the analysis of existing chemicals for which no monitoring data exist or when available monitoring data are not applicable to a particular exposure scenario.

Predictions of the expected airborne concentration of a substance may be based on monitoring data for analogous substances, OSHA PEL's for substances present in the workplace, and mass balance models.

a. Using Monitoring Data

When monitoring data on occupational exposure are available, the utility of the data should be evaluated following the process established for CEB that is described in "Guidelines for Statistical Analysis of Occupational Exposure Data" (PEI 1989). The quality of the exposure data that CEB assesses varies from poorly characterized, summary data to well-characterized sets of individual data points. In almost all instances, the quantity of available data is limited. The guidelines describe the treatment of uncertainties, assumptions, and biases in the data. With the assistance of an industrial hygienist and a statistician, the CEB engineer can use the guidelines to categorize the data and perform the statistical analysis. Appendix F briefly describes these guidelines.

b. Using Analogous Data

(1) General Methodology

The airborne concentration of vapors and particulates may be estimated using personal monitoring measurements for analogous chemicals or processes. In each case, similarities must exist in physical/chemical properties of the chemicals, nature of workplace environment, quantites of material handled, and worker activities associated with use of the chemical. Chapter III discusses typical sources of chemical-specific data, such as the OSHA Compliance Database, NIOSH industry-wide surveys, and NIOSH Health Hazard Evaluations (HHE's). In addition, CEB has developed descriptive generic scenarios that include summaries of the available monitoring data for industries that are frequently assessed. Currently, these scenarios cover metalworking, textile dyeing, and spray coating.

Although estimates of airborne concentrations may be based on analogous chemicals or processes, caution should be used when making the analogy. This is a type of reality check. If the exposure calculations exceed the PEL, Short-Term Exposure Level (STEL), or Immediately Dangerous to Life or Health (IDLH) levels, then the assumptions or calculation methods should be checked for their validity.

To estimate airborne concentrations for vapors from analogous data, the following simple relationship has been derived:

$$C_{v,s} = C_{v,k} \frac{P_s X_s}{P_k X_k} \qquad \text{Equation 4-4}$$

where:

$C_{v,s}$ = Estimated airborne concentration of the PMN chemical, ppm

$C_{v,k}$ = Measured airborne concentration of the known chemical, ppm

P_s = Vapor pressure of PMN chemical, torr

P_k = Vapor pressure of known chemical, torr

X_s = Mole fraction of PMN chemical in mixture

X_k = Mole fraction of known chemical in mixture

The derivation of this relationship assumes that:

- Vapor generation is driven by either evaporation from an open surface or the displacement of saturated vapors from a container.
- The liquid temperatures (T) and the mass transfer coefficients (K) of the PMN chemical and the known substance are similar.
- The workplace environments are similar: the ventilation rates (Q) and mixing factors (k) for the workplaces in which the PMN and the known chemical are handled are essentially the same; the quantities of materials handled are similar (for example, in an activity such as drumming, the volume of the container and the fill rate for the PMN and the known are equal).
- Raoult's Law is valid.

Assuming the ideal gas law, the airborne concentration expressed on a volume basis in units of ppm [C_V] can be converted to airborne concentration expressed on a mass basis in units of mg/m^3 [C_m] using the following equation:

$$C_m = C_v \frac{M}{V} \qquad \text{Equation 4-5}$$

where:
- C_m = Airborne concentration, mg/m^3
- C_v = Airborne concentration, ppm
- M = Molecular weight of the chemical of interest, g/g-mole
- V = Molar volume, liter/mole (use 24.45 liter/mole at 25° C and 760 mm Hg)

For particulates, the analogous data may be expressed as the airborne concentration of a particular substance or as total solids. To calculate the airborne concentration, assume that the composition of the airborne particulates is the same as the composition of the bulk material. A ratio of the weight fractions of each substance is used to calculate the airborne concentration of the PMN from concentration of the known chemical.

$$C_{m,s} = C_{m,k} \frac{Y_s}{Y_k} \qquad \text{Equation 4-6}$$

where:
- $C_{m,s}$ = Estimated airborne concentration of the PMN chemical, mg/m^3
- $C_{m,k}$ = Measured airborne concentration of the known chemical, mg/m^3
- Y_s = Weight fraction of PMN chemical in mixture
- Y_k = Weight fraction of known chemical in mixture

If the airborne concentration is measured as total solids, this equation becomes:

$$C_{m,s} = C_{m,x}\ Y_s' \qquad \text{Equation 4-7}$$

where: $C_{m,s}$ = Estimated airborne concentration of the PMN chemical, mg/m^3

$C_{m,x}$ = Measured airborne concentration of total dust, mg/m^3

Y_s' = Weight fraction of PMN chemical in the solids

When using an analogy for particulates, the following parameters should be judged to be similar: hygroscopicity, moisture content, density, particle shape, particle size and distribution, and static buildup potential. Whenever available, the engineer should obtain particle size information for the PMN chemical. This information may be presented in several different ways, from a complete particle size distribution to the limited identification of average size or percent of particulates above or below certain cutoff sizes. Depending on the type of data submitted, the particle size information may be used to more completely characterize the exposure to the worker. The data should represent the PMN particle size distribution at the potential exposure points.

If properly collected and analyzed, particle size information may be used to identify the percent of particles in the respirable range. Respirable particulates are defined as those with an aerodynamic diameter of 3.5 μm or less. These particles are expected to reach the alveolated gas exchange portions of the human respiratory system where they may be absorbed. Almost all particulates that are inhaled and are larger than the respirable size are deposited in the upper respiratory tract. Those which deposit in the nasopharanx behind the nasal hairs tend to be carried downward to the throat. Those which deposit in traechiobronchial system are carried upward to epiglottis. Those particles that are larger than the respirable particles tend to be ingested. A CEB industrial hygienist should be consulted to determine the appropriate type of testing protocol for particle size testing, and to provide guidance to the submitter.

(2) <u>Spray Coating</u>

One use scenario which CEB frequently assesses is industrial spray application of coatings. To standardize and facilitate the engineering analysis, CEB developed a report entitled "Generic Engineering Assessment Spray Coating" (CEB 1987a) that describes several generic spray coating scenarios. The report, which is included in Volume II, is organized according to the industry of application: automotive finishing and refinishing, wood and metal furniture finishing, large appliance finishing, railroad car finishing, light aircraft finishing, and heavy machinery finishing. For each industry, a matrix provides information that can be used to estimate typical usage rate of coating, number of use sites, numbers of workers exposed by activity, duration and frequency of exposure, types of protective equipment likely to be used, and inhalation exposures to total mist and organic solvents.

Of the PMN chemicals that are intended for use in coatings, most are nonvolatile substances used as resins, pigments, or other additives in the coating. Therefore, the PMN chemical is usually part of the solids, and inhalation exposures can be estimated based on expected airborne concentrations of total mist using Equation 4-7. Table 4-2 lists airborne concentrations of total mist categorized by scenario.

TABLE 4-2. ESTIMATED AIRBORNE CONCENTRATIONS OF TOTAL MIST FOR SPRAY COATING OPERATIONS

Industry	Estimated 8-hr TWA (mg/m^3) Total Mist
Automotive	
- Finishing	Not expected to exceed levels for refinishing operations.
- Refinishing	5
Furniture	
- Wood	0.1 to 2.5
- Metal	0.1 to 23.5
Large appliance	35
Railroad car	<15
Light aircraft	<15
Heavy machinery	1 to 18

Source: O'Brien 1981, CEB 1987a

When no particular scenario represents the PMN use, the OSHA PEL for particulates, not otherwise regulated, should be used to represent the upper bound for potential exposure. Estimation of exposure based on PELs is discussed later.

If the PMN is volatile, the potential airborne concentration can be based on the airborne concentration of a known similar volatile compound. Airborne concentration levels for a variety of known solvents are listed in the NIOSH technical report entitled "Evaluation of Engineering Control Technology for Spray Painting" (O'Brien 1981). If most of the volatile portion of the coating becomes airborne during spray application, the airborne concentration of the PMN is estimated to be a function of the ratio of mole fractions:

$$C_{v,s} = C_{v,k} \frac{Y_s}{Y_k} \qquad \text{Equation 4-8}$$

where:
$C_{v,s}$ = Estimated airborne concentration of the PMN chemical, ppm
$C_{v,k}$ = Measured airborne concentration of the known chemical, ppm
Y_s = Weight fraction of PMN chemical in mixture
Y_k = Weight fraction of known chemical in mixture

(3) Textile Dye Weighers

A description of textile dyeing operations is found in the in-house report entitled "The Dyeing and Printing of Textile Fibers" (Heath 1984). This comprehensive overview includes information on types of dyes, typical formulations of dyes, days of operation, numbers of workers, throughput rates, and releases for both batch and continuous dyeing operations. Based on the report, dyes are used that range from less than 0.001 percent to greater than 4 percent fiber active coloring material based on the weight of the fiber (wof). The level of dye as formulated will depend on the depth of color needed. For example, the active coloring agent may be 1 percent of the dye solution if used as a moderate-deep base color and 0.1 percent if used as a shading agent. Darker shades such as black, navy blue, and orange may require 5 percent or more dye as formulated.

A typical dyehouse employs one dye weigher per shift and may operate 3 shifts per day over 5 to 7 days per week. Inhalation exposures to particulates during dye weighing are expected.

To better assess occupational exposures to powder dyes, EPA has conducted a study of dye weighers at 24 textile dye/print houses. Quantities of individual formulated dyes handled by workers in the study using a "scoop/shake/pour" method ranged from 0.001 to 54 kg/worker/shift. On an interim basis, CEB estimates the worst case and average dye dust concentration per weighing ratio as a basis for assessing workplace inhalation dust exposures. The assessment of worker exposure to dust via the inhalation route is calculated from the following formula:

Typical Case:

$$I = 0.0314 \times \text{\% concentration, based on solids} \times \text{no. weighings/day} \qquad \text{Equation 4-9}$$

Worst Case:

$$I = 0.170 \times \text{\% concentration based on solids} \times \text{no. weighings/day} \qquad \text{Equation 4-10}$$

The percent concentration is either provided in the PMN or other source or may be estimated (see Appendix C of Heath 1984). The number of weighings of dyestuff is an engineering judgement based on mass used per site and conditions of use. Table 4-3 provides typical values.

TABLE 4-3. TYPICAL PARAMETERS FOR DYE WEIGHING OPERATIONS

Type of operation	No. of dye units per site-day	No. of dyelots per shift/unit	No. of dyeings per dyelot
Batch	3	1.33	1-3
Continuous	1	1	3-4
Print	1	1	2-4/screen

Source: Heath 1984

(4) Weighing and Transfers of Small Quantities of Solids

Workers who weigh or transfer chemicals similarly to those in dye weinghing activities (i.e., similar quantities of less than 54 kg/worker/shift and similar handling techniques of "scoop/shake/pour") are expected to be exposed to similar levels of dust. For example, airborne concentration levels of paper and leather dyes have been found to be similar to levels of textile dyes (Heath 1988). Thus, in these instances, the equations for dye weighers can be used to estimate exposures.

(5) Metal Working Operations

Information on the operations, numbers of workers and activities, and occupational exposures at machine shops and selected metalworking operations can be found in "Exposure to N-Nitrosodiethanolamine in Machine Shops" (CEB 1984b) and "Exposure to N-Nitrosodiethanolamine in Selected Metalworking Operations" (CEB 1984c). Table 4-4 presents estimated inhalation exposures based on NIOSH field studies and OSHA compliance monitoring data collected for n-nitrosodiethanolamine (NDELA) and oil mist during the period of 1972 to 1984.

TABLE 4-4. ESTIMATED AIRBORNE CONCENTRATIONS FOR METALWORKING

Type of facility	Contaminant	Airborne concentration arithmetic average (range)
Machine shops	NDELA (vapor)[a]	0.04 $\mu g/m^3$ (nondetectable to 0.08 $\mu g/m^3$)
Machine shops	Oil mist	1.2 mg/m^3 (0.001 to 5 mg/m^3)[b]
Rolling mill	Oil mist	0.24 mg/m^3 (0.18 to 0.3 mg/m^3)

[a] N-Nitrosodiethanolamine (NDELA)
CAS: 1116-54-7
MW: 110
VP: not available

[b] The upper end of this range based on OSHA monitoring data is actually 8.3 mg/m^3 which exceeds the OSHA PEL for oil mist. Since exceedance of the PEL occurred only in a small number of inspections and the OSHA data may be biased by the fact that inspections are often conducted based on employees complaints about the workplace, the OSHA PEL for oil mist has been used to represent the upper bound.

Source: CEB 1984b, CEB 1984c

The OSHA PEL for oil mist should be used to represent worst case exposures for metalworking operations. Use of OSHA PEL's as surrogates for exposure or as upper bounds is discussed later.

(6) Printing

In assessing inhalation exposures in the printing industry, CEB relies on a very limited data set. High-speed letterpress and lithographic printing operations are expected to generate ink mist. High-speed processes include printing of newspaper (300 to 400 m/min) and publications (300 to 370 m/min). Based on two studies, occupational exposures to ink mist during printing operations range from 0.3 to 6.2 mg/m^3 (Gikis 1983). Occupational exposures to solvents during printing operations are also expected.

(7) Tire Manufacturing

Air monitoring data for occupational exposures to total particulates are available for tire manufacturing (SIC code 3011). From 1979 to 1980, NIOSH conducted detailed control technology assessments of seven tire manufacturing sites. The individual plant reports contain air sampling data, identify emission sources, describe work practices, and evaluate engineering controls for various operations such as weighing, mixing, milling, and curing. The results of these field studies are summarized in the NIOSH report entitled "Control of Air Contaminants in Tire Manufacture" (NIOSH 1984).

The NIOSH studies found that workers may be exposed to particulates from rubber, carbon black, process oils, vulcanizing agents, accelerators, activators pigments, softeners, and plasticizers. Table 4-5 gives a typical compound for tire manufacture.

TABLE 4-5. TYPICAL COMPOUND COMPOSITION FOR TIRE MANUFACTURE

Compound	wt %
Rubber	71
Carbon black	18
Zinc oxide	2
Stearic acid	1.4
Softener	3.4
Antioxidant	0.7
Sulfur	2.0
Primary accelerator	0.5
Secondary accelerator	0.1

Source: NIOSH 1984

Workers may be exposed to dusts during transfers, weighing, and mixing of raw materials. During milling, extrusion, calendering, and curing, workers may be exposed to fumes that are emitted from the hot milled rubber. During tire repair, workers are potentially exposed to rubber particles from grinding. Exposures are generally higher in the earlier stages of the manufacturing process when dry ingredients are handled than in the latter stages that involve the building and handling of cured product. The NIOSH report describes the operations and worker activities in detail.

PMN chemicals are frequently nonvolatile substances used as resins, pigments, or other additives in rubber compounds. Therefore, the PMN chemical is usually part of the solids and inhalation exposures can be estimated based on expected airborne concentrations of total solids using Equation 4-7.

Table 4-6 lists airborne concentrations of total particulates categorized by worker activity based on the NIOSH field studies. In engineering assessments for tire manufacturing, note that carbon black has an OSHA PEL of 3.5 mg/m^3, 8-hr TWA.

An earlier study found that particulate levels in compounding and mixing are about 1 to 3 mg/m^3, while those in milling, curing, and finishing are generally less than 1 mg/m^3 (Williams 1980). These results agree with the NIOSH data presented in Table 4-6.

TABLE 4-6. SUMMARY OF NIOSH MONITORING DATA FOR PARTICULATE EXPOSURES IN TIRE-MANUFACTURING OPERATIONS BASED ON SEVEN PLANTS

Activity	Total particulates range of exposures (mg/m^3)
Bin-filling (emptying bags into bins)	0.6 to 4.3
Manual weighing with LEV	1.5 to 2.2
Manual weighing without LEV	2.1 to 2.5
Mixing	0.08 to 1.54
Milling with LEV	0.2 to 1.22
Calendaring (under canopy hood)	0.07 to 0.4
Curing (automatic press/general ventilation)	0.1 to 0.22
Tire repair with LEV	0.17 to 0.26

Source: NIOSH 1984

If the PMN is volatile, the potential airborne concentration can be based on the airborne concentration of a known similar volatile compound using Equation 4-4.

Worker exposure to pentane, hexane, heptane, benzene, and toluene during tire manufacture has been studied (Van Ert 1980). Since PMN chemicals typically are non-volatile, the solvent exposure data are not presented here.

c. Use of OSHA PEL's

When neither representative monitoring data nor a generic scenario are available to describe potential exposures to the chemical of interest, the upper bound to the airborne concentration may be estimated using OSHA PELs for analogous substances or substances present in the same workplace as the PMN chemical. To estimate the airborne concentration, the PEL is used in the same equations as those used to calculate concentrations from measured data.

The most frequent use of PELs is in the estimation of airborne concentrations of solids for operations such as spray coating or solids handling. The OSHA PEL for particulate, not elsewhere regulated, is 10 mg/m^3, 8-hr TWA (29 CFR 1910.100, Table Z-1-A). Data collected by OSHA and NIOSH for total dust exposures have been tabulated by SIC code (Mitre 1985). A mean exposure value was calculated for each SIC code. The mean exposures support the use of 15 mg/m^3 to represent airborne concentrations of total dust in the worker's breathing zone as a substitute for specific data. This study has been included in Volume II.

The OSHA PEL is frequently used to describe exposures to oil mists generated in operations such as metalworking and printing presses. The 8-hr TWA PEL for oil mists is 5 mg/m^3 (29 CFR 1910.100, Table Z-1-A).

To calculate the airborne concentration of PMN from either of these PEL's, the following variant of Equation 4-7 is used:

$$C_{m,s} = KC_k Y_s' \qquad \text{Equation 4-11}$$

where:

$C_{m,s}$ = Estimated airborne concentration of PMN substance

KC_k = 8-hr TWA airborne concentration of known (either particulate, not elsewhere regulated or oil), mg/m^3

Y_s' = Weight fraction of PMN chemical in the solids or oil

To calculate the airborne concentration for vapors based on the PEL, use Equation 4-4 substituting the PEL concentration (in ppm) for the measured concentration.

d. Use of Mass Balance Models

When information on analogous chemicals is not available, CEB engineers use mass-balance or "box" models to predict airborne concentrations. These models are most applicable for vapor and gaseous emissions because vapors follow currents freely and are not influenced by gravity.

Worker exposures can be influenced by many variables: 1) degree of automation, 2) employee work practices, 3) equipment design, age, and frequency of maintenance, 4) container and closure design, 5) ventilation type and rate, 6) employee use of protective equipment, 7) effectiveness of emission control devices, and 8) product flammability. These factors must be considered before estimating worker exposure using these models. This is especially true if the default parameters are used to represent worst and typical conditions in the workplace.

One check of the models is that good design of equipment keeps concentrations below 25 percent of the Lower Explosive Limit (LEL). A reasonable worst case assumption should not exceed good design practice. For PMN chemicals, however, the LEL or operating practices may not be known.

(1) Simple Mathematical Model

The model frequently used to estimate worker exposures is based on the mass balance of a substance in an enclosed space. It assumes the following:

- There is perfect and instantaneous mixing of the contaminant with incoming general dilution air.
- The airborne concentration of the contaminant in the exfiltration air is the same as in the room.
- Only one source within the work area emits the contaminant.

The basic mass balance is expressed as:

$$V\frac{dC_v}{dt} = QC_1 + S - \left[\left(\frac{Q}{V} + e\right) VC_v\right] \qquad \text{Equation 4-12}$$

where:
- Cv = Contaminant concentration in workplace, [volume chemical/volume air]
- Q = Volumetric ventilation rate [volume/time]
- C = Contaminant concentration in incoming dilution (infiltration) air [volume/volume]
- S = Source generation rate (volumetric) [volume/time]
- e = Extinction rate [$time^{-1}$]
- V = Room volume [volume]

The following assumptions are made to simplify the model:

- Extinction of the chemical (adsorption, absorption, or chemical transformation) resulting from deposition on walls and equipment, condensation of hot vapors, and photodegradation of chemicals is negligible.

- The incoming air is contaminant-free.
- The concentration of contaminant at initial time (t=0) is negligible.
- The generation and ventilation rates are constant over time.
- Room air and ventilation air mix ideally.
- The concentration approaches the equilibrium concentration.

In its most simplified form, the model is expressed as:

$$C_v = \frac{S}{Q} \qquad \text{Equation 4-13}$$

where: C_v = Contaminant concentration in workplace, ppm
S = Source generation rate (volumetric), cm^3/h
Q = Volumetric ventilation rate, cm^3/h

If available, a known ventilation rate for the workplace under study is used. Volumetric ventilation rates are conventionally represented by units of ft^3/min or cfm. General ventilation rates in industry range from a low of 500 ft^3/min to over 10,000 ft^3/min; a typical value is 3,000 ft^3/min (Clement 1982). If the ventilation rate is not known, a rate of 3,000 cfm (85 m^3/min) is considered typical and 500 cfm (14.2 m^3/min) represents worst case. For outdoor operations with only minimal structure, the ventilation rate in cfm is estimated as 26,400 v where v is the wind speed in mph (Clement 1982). The average wind velocity is assumed to be 9 mph (Clement 1982).

In reality, general dilution ventilation air does not always mix perfectly and instantaneously with contaminated room air. Pockets of poorly mixed air may be found in the room. Thus, a dimensionless mixing factor (k) is introduced to describe the degree of mixing of the displaced ventilated air. Ideal mixing is represented by a mixing factor (k) of 1. This mixing factor is a function of room size and locations of the air inlet and exhaust. The ACGIH Ventilation Handbook suggests the following factors (ACGIH 1988): best (0.67 to 1), good (0.5 to 0.67), fair (0.2 to 0.5), and poor (0.1 to 0.2).

In using this model, a mixing factor of 0.5 represents a typical case and 0.1 represents a worst case. Incorporating the mixing factor and assuming the ideal gas law, the model is expressed as:

$$C_v = \frac{(1.7 \times 10^5)\ T_a\ G}{M\ \ Q\ \ k} \qquad \text{Equation 4-14}$$

where:
- C_v = Contaminant concentration in workplace, ppm
- T_a = Ambient temperature of the air, K
- G = Vapor generation rate, g/sec
- M = Molecular weight, g/g-mole
- Q = Ventilation rate, ft^3/min or cfm (Note that "cfm" is a conventional unit for volumetric ventilation rates)
- k = Mixing factor, dimensionless

Note: the factor 1.7×10^5 in Equation 4-14 accounts for units conversion and is expressed in units of:

$$\frac{sec\ atm\ cm^3\ h\ ft^3}{gmol\ K\ \mathrm{min}\ cm^3}$$

Equation 4-14 is the model which CEB engineers frequently use to estimate the airborne concentrations of contaminants in the workplace. Use of this model for specific scenarios will be discussed later.

(2) <u>Complex Mass Balance Models</u>

The basic mass balance approach can be used to derive models for other scenarios (e.g., when the infiltration air is not contaminant-free, the concentration is not constant, or the initial concentration of contaminant is known). There are three complex mass balance models that have previously been used by CEB to estimate room concentrations. These are the Multi-Chamber Chemical Exposure Model (MCCEM), the Consumer Products Safety Commission (CPSC) model, and the Monte Carlo Simulation model. They have similar mass balance approaches to model exposures to chemicals.

The MCCEM was developed for EPA, Office of Research and Development (ORD) in Las Vegas, Nevada. MCCEM is available on diskette and estimates airborne concentrations in the home based on user-supplied generation rates (Geomet 1989). The model provides default values for room size, house size, infiltration rates, and interzonal air flows. CEB has used this model to estimate occupational exposures to house painters when using latex paints containing formaldehyde-releasing biocides. Further information about the model is available in Volume III.

The CPSC model calculates exposure concentrations resulting from a continuous release of a chemical substance in an enclosed space. This model has been used by CEB to estimate exposure to chlorinated solvents. If the release of the PMN chemical occurs over a relatively short time but the worker remains in the area for other activities throughout the period, the mass balance equation must be solved for two distinct periods. The first period is the time when the PMN chemical is released (t_0) until the time the PMN chemical has ceased being released (t_1). The second time period is after the PMN chemical has ceased being released (t_1) until the time the worker leaves the room (t_2). Other assumptions for the model include that the pollutant concentration at the start of the release, the outdoor pollutant concentration, and the rate of removal by extinction are all zero. The model also assumes that the source strength and air changes per hour are constant over the time of interest. The model also assumes ideal mixing. The general solution for estimating room concentrations during the two time periods is presented in Equation 4-15.

$$C(t_0 \text{ to } t_2) = \frac{S\ t_1}{a\ V} - \frac{S}{a^2 V}\,[1 - e^{-a(t_1 - t_0)}][e^{-a(t_2 - t_1)}] \quad \text{Equation 4-15}$$

where: $C(t)$ = Contaminant concentration in the workplace, ppm

S = Source strength, cm^3/h

a = Number of air exchanges per hour for the ventilation system, h^{-1}

V = Room volume, m^3

t_0 = Time when the pollutant is released, h

t_1 = Time when the pollutant release ceases, h

t_2 = Time when the worker leaves the room, h

The Monte Carlo Simulation Model is a combination of a complex mass balance model to estimate exposure concentrations of the chemical substance in an enclosed space and a Monte Carlo analysis. The continuous release model estimates the concentration of a chemical substance that is released at a constant rate over a period of time until the exposure duration ends or the source ceases to emit a substance, whichever comes first. This model has been computerized by EPA, ORD. This model has been used to estimate room concentrations after releases of chlorofluorocarbons in various scenarios such as mobile air conditioning and refrigeration. Assumptions for this model include that the pollutant concentration at the start of the release, the outdoor pollutant concentration, and the rate of removal by extinction are zero. The model contains a factor that accounts for nonideal mixing. The solution of the mass balance model for this model is presented in Equation 4-16.

$$C = \left(\frac{Sr_t}{Vm} + C_o\right)\left(1 - e^{-\left(\frac{m}{r_t}\right)t}\right) + C_b e^{-\left(\frac{m}{r_t}\right)t} \qquad \text{Equation 4-16}$$

where:

- C = Contaminant concentration in the workplace, g/m^3
- S = Source generation rate, g/h
- r_t = Residence time of the indoor air, h
- m = Mixing factor
- V = Room volume, m
- C_o = Outdoor concentration, g/m^3
- C_b = Concentration in the room at the beginning, g/m^3
- t = Time, h

The Monte Carlo analysis portion of this model consists of three major steps. First, the CEB engineer selects a mean, standard deviation, minimum, maximum, and distribution type for each of the input variables in Equation 4-16. Second, the model is executed numerous times (usually between 500 and 1000 times), each time with a unique combination of values for the input factors. The unique sets of input factors are generated by the Monte Carlo model randomly sampling from the distribution assigned to the factors. Third, the numerous values of exposure generated from this iteration technique are then analyzed statistically to estimate a mean, standard deviation, minimum and maximum, and 95th percentile. The primary statistic is the mean value for each scenario. This value is more appropriately referred to as the estimated likely exposure because it is derived using a variation in input values. In most cases, the predicted 95th percentile value may be used to represent the reasonable worst-case exposure. The validity of this model is highly dependent on how well the distribution of each input parameter is understood. Interpretations of the results must be made very cautiously if uncertainty exists in the understanding of the input distributions.

(3) Other approaches

Other approaches that are not commonly used are included here for general information. The mass balance modeling approach does not account for dispersion patterns, room size, or location of the worker with respect to the source. To determine the effect of worker location on inhalation exposures, data were collected during laboratory pilot studies and an empirical model for predicting airborne concentrations during drumming was developed (MRI 1986). This model incorporates a factor to account for the positioning of a motionless worker in relation to the source and the ventilation flow direction. This distribution factor (d) represents the amount of time a worker spends in front of the fill station. The model does not incorporate the effect of worker movement because standardization of worker movement is nearly impossible. Slight movement may reduce the breathing height that may affect concentrations. The model is expressed as:

$$C_m = \left[\frac{2.1 \times 10^6}{Q} + (1000d)\right] G \qquad \text{Equation 4-17}$$

where:
- C_m = Airborne concentration, mg/m^3
- Q = Ventilation rate, ft^3/min
- d = Distribution factor, sec/m^3
- G = Vapor generation rate, g/sec

The final MRI report provides the following table of default distribution factors for drum filling operations of liquids. One conclusion drawn from the study is that cross ventilation air flow produced lower exposures than rear ventilation air flow. CEB has not used this model to date in preparing assessments.

TABLE 4-7. DEFAULT DISTRIBUTION FACTORS

		d factors (sec/m^3)	
Filling operations	Airflow	MW < 100	MW > 100
Automated	Side	0.2	0.1
	Rear	0.6	0.6
Manual	Side	0.3	0.1
	Rear	3	3

Source: MRI 1986.

(4) Modeling the Generation Rates

To estimate the airborne concentration of the contaminant, the generation rate of the contaminant must be estimated. Several approaches to calculating the generation rate have been developed.

Many models which CEB routinely uses to estimate the generation rate are functions of vapor pressure of the substance or mixture. The vapor pressure of a pure substance may be available in the literature. ICB usually provides estimates of vapor pressure for PMN chemicals based on the chemical structure.

If two boiling points at two different temperatures (T_1, T_2) and pressures (P_1,P_2) are known, the vapor pressure at 25° C (P_{25}) may be calculated as shown in equation 4-18:

$$P^{\bullet}_{25} = P_1 \left(\frac{P_2}{P_1}\right)^{b} \qquad \text{Equation 4-18}$$

$$[P] = \text{atm}$$

where:

$$b = \frac{T_2\,(298 - T_1)}{298(T_2 - T_1)}$$

$$[T] = \text{Kelvin} = \text{K}$$

Equation 4-16 is derived from the Clausius-Clapeyron Equation, and assumes that the heat of vaporization, $^{\bullet}H_m$, is constant. If the heat of vaporization and one boiling point (T_1 at P_1) are known, the Clausius-Clapeyron equation may be used to calculate the vapor pressure. This is presented in Equation 4-19.

$$ln\left(\frac{P^{\bullet}_{25}}{P_1}\right) = \frac{^{\bullet}H_m}{R}\left(\frac{1}{T_1} - \frac{1}{298}\right) \qquad \text{Equation 4-19}$$

where: R = Universal gas constant, 8.314 J/mol $\cdot$K
$^{\bullet}H_m$ = Heat of vaporization, J/mol

For mixtures, the vapor pressure of a component (P_a in atmospheres) may be calculated using Raoult's Law as shown in Equation 4-20:

$$P_a = P^* X_a \qquad \text{Equation 4-20}$$

where:

$$X_a = \frac{\frac{W_a}{M_a}}{\frac{W_a}{M_a} + \frac{W_b}{M_b} + \ldots \frac{W_i}{M_i}}$$

P^* = Vapor pressure of pure substance, atm

X = Mole fraction of component

W_i = Weight percent of component

M_i = Molecular weight of component, g/g-mole

(a,b,i denote components)

Raoult's Law may be too simplistic in certain circumstances because vapor/liquid equilibria data do exist for certain binary systems. Where data exist, this should be used instead of calculated values. For chemicals that are solid at ambient temperature and volatilize by sublimation, standard chemical engineering techniques for estimating physical properties are not adequate for predicting vapor pressure.

Transfer Operations. When liquids are transferred, displacement of saturated vapor from the container must be considered. If evaporation rate is negligible in comparison to the displacement rate, the generation rate is expressed as:

$$G = \frac{f\ M\ V\ r\ P^{\bullet}}{3600\ R\ T_L} \qquad \text{Equation 4-21}$$

where:
- G = Vapor generation rate, g/sec
- f = Saturation factor, dimensionless
- M = Molecular weight, g/g-mol
- V = Volume of container, cm^3
- r = Fill rate, units/hr
- $P^{\bullet}$ = Vapor pressure of pure substance, atm
- R = Universal gas constant, 82.05 atm cm^3/g-mol °K
- T_L = Liquid temperature, °K

Equation 4-21 is frequently used to estimate worker exposures during tank truck or tank car loading and drumming operations. Summary Table 4-11 at the end of Section 4A presents the default values generally assumed by CEB for container volume and fill rate. If complete saturation of the vapor space within the vessel is assumed, the saturation factor is equal to 1.

This model is used to estimate losses due to vapors generated from bulk loading of petroleum products as described in AP-42 (USEPA 1985b). Saturation factors for tank truck loading of petroleum liquids are expected to range from 0.5 to 1.45 (USEPA 1985b). Table 4-8 lists typical saturation factors by mode of loading for tank trucks and tank cars.

TABLE 4-8. SATURATION FACTORS FOR BULK LOADING OPERATION

Mode of operation	Saturation Factor (f), dimensionless
Submerged loading:	
Clean cargo vessel	0.50
Normal dedicated service	0.60
Dedicated vapor balance service	1.00
Splash loading	
Clean cargo vessel	1.45
Normal dedicated service	1.45
Dedicated vapor balance service	1.00

Source: U.S. EPA 1985b.

CEB engineers typically assume complete saturation of the vapor space within the vessel ($f = 1$) for both tank truck and tank car loading. Typical and worst case parameters for tank truck and tank car loading are presented in summary Table 4-11 at the end of Section 4A.

An alternative method for estimating the vapor generation rate for bulk loading of tank trucks and cars is the use of emission factors. OAQPS has collected information on air emissions of volatile chemicals from different industrial processes. These data are compiled into a report entitled "Toxic Air Pollutant Emission Factors - A Compilation for Selected Air Toxic Compounds and Sources" (USEPA 1988a). Emission factors from tank truck and car loading expressed as amount of chemical released per tank truck/car loaded are presented for several chemicals. Table 4-9 presents estimated generation rates assuming that two tank trucks or one tank car can be loaded in an hour.

TABLE 4-9. AIR EMISSION FACTORS FOR LOADING

Operation/chemical	Generation rate (g/sec)
Tank truck loading	
ethylene dibromide (product)	1×10^{-6}
ethylene dibromide (gasoline)	1×10^{-3}
ethylene dichloride (product)	1×10^{-5}
ethylene dichloride (gasoline)	2.6×10^{-3}
benzene	0.06
carbon tetrachloride, controlled	0.4
carbon tetrachloride, uncontrolled	2.7
chloroform	3.9
Tank car loading	
acrylonitrile, wet scrubber	4.72×10^{-6}
acrylonitrile	4.6×10^{-4}

Source: USEPA 1988a.

These generation rates were compared to the model (Equation 4-21). The model was found generally to overestimate the generation rate by several orders of magnitude. This overestimation is likely because the model does not consider the effects of engineering controls used to recover vapor losses as do the actual data.

Two modes of drum-filling were studied under two simulated drum-filling conditions: 1) splash filling in which the liquid dispenser remains at the top of the container and the liquid splashes freely, and 2) bottom filling in which the liquid dispenser remains at the bottom of the drum to minimize volatilization (MRI 1986). During splash filling, the saturation concentration was reached or exceeded by misting (MRI 1986). The generation rate for bottom filling was one-half that for splash filling. Thus, for bottom filling of drums, the saturation factor is expected to be about 0.5.

Open Surfaces. Open surface operations include work related to open vats or tanks, solvent dip tanks, open roller coating, and cleaning or maintenance activities. Although this model is frequently used to estimate air emissions from spills, CEB typically estimates exposures only during routine operations.

Under contract to EPA, the evaporation rates of different pure compounds in a test chamber were measured to determine an empirical model to describe the relationship between evaporation rate and physical chemical properties. The experiment is described in "Evaporation Rate of Volatile Liquids" (Pace Laboratories 1989). The data are presented in this reference for 16 compounds studied at different air velocities and temperatures. The data were curve-fitted. CEB is especially concerned about low volatility chemicals and low air flow rates.

Based on mass balance of a differential element above a liquid pool, the evaporation rate was derived (see Appendix K).

$$G = \frac{13.3167\ M\ P^{o}\ A}{T} \left[\frac{D_{ab}\ v_z}{\Delta z}\right]^{0.5} \qquad \text{Equation 4-22}$$

where:
- G = Generation rate, lb/hr
- M = Molecular weight, lb/lb mole
- P^o = Vapor pressure, in. Hg
- A = Area, ft^2
- D_{ab} = Diffusion coefficient, ft^2/sec of a through b (in this case b is air)
- v_z = Air velocity, ft/min
- T = Temperature, °K
- Δz = Pool length along flow direction, ft

Gas diffusivities of volatile compounds in air are available for several existing chemicals. However, the diffusion coefficient often will not be known. An equation to estimate diffusion coefficients has been developed (see Appendix K). The expression for the diffusion coefficient is expressed as:

$$D_{ab} = \frac{4.09 \times 10^{-5} (T)^{1.9} \left[\frac{1}{29} + \frac{1}{M}\right]^{0.5} (M)^{-0.33}}{P_t} \qquad \text{Equation 4-23}$$

where:
- D_{ab} = Diffusion coefficient, cm^2/sec
- T = Temperature, °K
- M = Molecular weight, g/g-mole
- P_t = Pressure, atm

Substituting into evap:

$$G = \frac{8.24 \times 10^{-8}\ M^{0.835}\ P^{o} \left(\dfrac{1}{29} + \dfrac{1}{M}\right)^{0.25} (v_z)^{0.5}\ A}{T^{0.05} \Delta z^{0.5}\ P_t^{0.5}} \qquad \text{Equation 4-24}$$

where:

G	=	Generation rate, g/sec
M	=	Molecular weight, g/g-mole
P^o	=	Vapor pressure, mm Hg
v_z	=	Air velocity, ft/min
A	=	Area, cm^2
T	=	Temperature, °K
Δz	=	Pool length along flow direction, cm
P_t	=	Overall pressure, atm

This equation was tested by comparing it to experimental data provided by Pace Laboratories. In an extensive study using a specially-built apparatus, Pace measured the evaporation rate of 15 different compounds at several different temperatures and air velocities, and fit the data against "power law" regression against molecular weight, vapor pressure and air velocity, with generally good results. They performed an overall regression analysis for all chemicals except the "low vapor pressure" alcohols (1-hexanol, 1-heptanol, and 2-octanol) and obtained the following equation:

$$G = 0.000237(M)(P^{o})(v_z^{0.825}) \qquad \text{Equation 4-25}$$

where:

G	=	Generation rate, lb/hr ft^2
M	=	Molecular weight of evaporating substance, lb/lb mole
P^o	=	Vapor pressure at liquid temperature, in.Hg
v_z	=	Air velocity, ft/min

Because open surface operations are highly process-specific, the engineer must make a judgment about the surface area from which volatilization is occurring. Table 4-10 gives conversions for typical diameter openings and surface areas:

TABLE 4-10. TYPICAL DIAMETERS AND AREAS

Diameter	z, cm	Surface area, cm^2
3 ft	91.5	6,500
2 ft	61	3,000
1 ft	30.5	700
6 in	15.25	180
3 in	7.6	42
2 in	5.08	20
1 in	2.54	5

Sampling. Sampling methods vary and include dipping, open loop and closed loop sample bombs. For worst case assumptions, sampling from a tap into an open container should be assumed. Equation 4-24 is used assuming that the surface area is equivalent to the sample bottle opening. When no other information is available, estimate the surface area to be 80 cm^2 and 40 cm^2 for the worst case and typical case, respectively.

Fugitive Releases. Fugitive releases from valves, pump seals, and connections can lead to significant occupational exposures, especially in enclosed areas. Potential exposures are expected to be much less outdoors. To estimate occupational exposure from fugitive releases of gas or vapor for a new chemical from equipment used in its manufacture, the number of fugitive sources and the duration of worker exposure must be known. This information is typically not available at the PMN review stage. OTS rarely estimates occupational exposure from fugitive emissions of PMN chemicals during the review process since it is assumed that fugitive releases are not the most significant sources of occupational exposure.

The amount of contaminant leaking into the atmosphere can be estimated based on emission factors for the different equipment components (see

Table 6-1). The vapor generation rate is calculated as a function of the number of components, the appropriate emission factor for the component, and the weight fraction of the component in the process stream. These emission factors were developed for air pollution purposes and are discussed in Section VI of this manual.

(5) Summary Tables

The appropriate vapor generation rate is substituted into the mass balance model to calculate the airborne concentration of the pollutant. The worst case workplace conditions are represented by low general ventilation rates; typical conditions are represented by greater ventilation rates. If the workplace is known to have different ventilation rates than those generally assumed, the documented values should be used as the basis for estimation.

To facilitate calculations of exposure, Tables 4-11 and 4-12 list the typical and worst case assumptions for each potential source of exposure and present the simplified concentration model. The generation rates are based on equation 4-21 for transfers and 4-24 for open surfaces. The engineer should consider the applicability of the parameters before using these models.

TABLE 4-11. SUMMARY OF CONCENTRATION CALCULATIONS FOR TRANSFER OPERATIONS

	f	V cm^3	r h^{-1}	Q ft^3/min	k	C_v = 0.000756 VPfr/Qk ppm
Drumming (55 gal).						
Worst case	1	2.1×10^5	30	500	0.1	95 P
Typical case	0.5	2.1×10^5	20	3,000	0.5	P
Cans/bottles (5 gal.)						
Worst case	1	1.9×10^4	30	500	0.1	8.6 P
Typical case	0.5	1.9×10^4	20	3,000	0.5	0.1 P
Tank truck (5,000 gal.)						
Worst case	1	1.9×10^7	2	26,400 v[b]	0.1	11 P/v
Typical case	1	1.9×10^7	2	237,600	0.5	0.24 P
Tank car (20,000 gal.)						
Worst case	1	7.6×10^7	1	26,400 v[b]	0.1	22 P/v
Typical case	1	7.6×10^7	1	237,600	0.5	0.48 P

[a] The units for P are in mm Hg.

[b] Worst case outdoor ventilation (lowest wind speed, v in mph) should be estimated by CEB engineer.

TABLE 4-12. SUMMARY OF CONCENTRATION CALCULATIONS FOR OPEN SURFACES

	A cm^2	z cm	Q ft^3/min	k	$C_v = 31.4\,P\,(1/29 + 1/M)^{0.25}\,A/M^{0.165}z^{0.5}Qk$ ppm[a]
Sampling					
Worst case	80	10	500	0.1	$16\,P\,(1/29 + 1/M)^{0.25}/M^{0.165}$
Typical case	40	7	3,500	0.5	$0.3\,P\,(1/29 + 1/M)^{0.25}/M^{0.165}$
Open surface					
Worst case	b	b	500	0.1	$0.628\,P\,(1/29 + 1/M)^{0.25}\,A/M^{0.165}z^{0.5}$
Typical case	b	b	3000	0.5	$0.021\,P\,(1/29 + 1/M)^{0.25}\,A/M^{0.165}z^{0.5}$

[a] The units for P are in mm Hg.

[b] See Table 4-10.

B. Estimating Dermal Exposure

1. General Information

In comparison to inhalation exposure, the assessment methodology for predicting dermal exposures is relatively simplistic. To assess the potential for dermal contact with a chemical, it is necessary to identify the activity where potential contact may occur, the likelihood of contact, the frequency of contact, the potential surface area of contact, the physical state of the contacted substance, associated chemicals, and the likelihood and effectiveness of the use of protective equipment. With this information, appropriate assumptions can be made to complete the assessment.

For liquids and many solids such as powders, granules, or flakes, a quantitative estimate of contact should be made. For materials such as gases, cast solids, or corrosives, a quantitative estimate should not be made. For these materials a qualitative estimate should be made using the following guidelines:

- Corrosives - Express contact as negligible due to the corrosive nature of the substance or associated compounds. Use for PMNs determined to be corrosive by SAT/HERD or for PMNs/mixtures with pH greater than 12 or less than 2. Consider contact points at which corrosivity may not apply, as in dilute solutions, and quantify for them as needed.

- High temperatures - Express contact as negligible for materials that are at temperatures above 140°F. Consider contact points at which temperature would not be a factor.
- Cast solids/PMN in matrices - Do not quantify contact. If the material is manually transferred, acknowledge that some surface contact may occur.
- "Dry" surface coatings (e.g., fiber spin finishes) - If manual handling is necessary and there is an indication that the material may abrade from the surface, quantify contact with fingers/palms as appropriate.
- Gases/vapors - Do not quantify contact, but acknowledge that some dermal contact will occur in the absence of protective clothing.

Qualitative estimates should describe dermal exposure using the following exposure categories:

- None - This is used to describe workers who have no chance of dermal contact during normal job activities.
- Very low - This is used to describe workers who during typical job activities would have no dermal exposure but who, on occasion, may have short periods of exposure after which all contact surfaces would be washed.
- Incidental contact - This is used to describe workers who during typical job activities have occasional dermal contact of a minor nature.
- Intermittent contact - This is used to describe workers who during typical job activities have dermal contact such as splashes, wiping with contaminated rags, contact with contaminated tools, or surfaces. Contact may be with either liquids or solids.
- Routine contact - This is used to describe workers who during typical job activities routinely have dermal contact not including immersion in a liquid.
- Routine immersion - This is used to describe workers who typically immerse their hand(s) into a liquid.

Worker practices and the use of dermal personal protective equipment are strongly dependent on the industry under study. Therefore, when making qualitative estimates it is important to obtain as much current information on work practices in the industry as possible.

2. Quantifying Dermal Contact

Dermal contact is best quantified with measurements made in specific operations in the industry under study. Techniques used to quantify dermal exposure include a patch type of dermal dosimeter made of gauze or charcoal cloth, a skin wash technique (most appropriate for chemicals with low rates of dermal absorption), urinary excretion of the chemical or metabolites of the chemical, and fluorescence of selective chemicals. It should be noted that these methods are commonly used to quantify exposure, but are very difficult to interpret (this science is still in formative stages). Since many of these techniques may be applied when the worker is wearing personal protective equipment it is important to find out the exact circumstances when the measurements were made.

Once it has been determined that dermal contact is likely and no monitoring data are available, the contact may be quantified by using Equation 4-26.

$$D = SQC \qquad \text{Equation 4-26}$$

where:
- D = Dermal exposure, mg
- S = Surface area of contact, cm^2
- Q = Quantity typically remaining on skin, mg/cm^2
- C = Concentration of chemical of concern, percent

The time of exposure is estimated qualitatively and the dermal exposure calculated using Equation 4-26 is then expressed as mg/day. It is very important to note the methodology used to determine the amount of dermal exposure.

Table 4-13 provides typical factors for estimating the amount of dermal contact that may occur in particular situations, if such contact is not ruled out by factors such as temperature or corrosivity. When using the typical values in

TABLE 4-13. TYPICAL FACTORS FOR CALCULATION OF DERMAL EXPOSURE

Activity	Typical examples	s[a] cm^2	Q[b] mg/cm^2	Resulting typical contact,[c] mg
Routine immersion, 2 hands	• Handling wet surfaces	1300	5-14	6500 to 18,200
	• Filling/dumping containers of powders, flakes, granules			
	• Spray painting			
Routine contact, 2 hands	• Maintenance/manual cleaning of equipment	1300	1-3	1300 to 3900
	• Unloading filter cake			
	• Changing filter			
	• Filling drums with liquid			
Routine contact, 1 hand	•	650	1-3	650 to 1950
Incidental contact, 2 hands	• Connecting transfer line	1300	1-3	1300 to 1900
	• Weighing powder/scooping/mixing (i.e., dye weighing)			
Incidental contact, 1 hand	• Sampling	650	1-3	650 to 1950
	• Ladling liquid/bench scale liquid transfer			

[a] Popendorf and Leffingwell 1982.

[b] Versar 1984.

[c] These estimates also must be adjusted by the concentration of the chemical in the mixture and the percent of the hand exposed if this is less than what would be typical. Concentrations that change over time due to evaporation or other factors also should be accounted for.

Table 4-13, the exposure estimate also should be adjusted by the following factors when applicable:

- The concentration of the chemical in a mixture;
- The percent of the hand exposed if less than what would be typically expected for the activity;
- Rapid evaporation of the chemical (lessening exposure time); and

- The effect of an industrial hygiene program.

The likelihood of the use of protective clothing and its effectiveness should be evaluated as described in Section IV.C. If gloves are worn for reasons other than strictly to protect against corrosivity, it must be assumed that there is some potential they will not always be used. In instances where gloves are expected, the engineer should clearly state the reasons for their use and estimate contact if they are not worn. The engineer must always clearly state the reasons for the conclusions.

The surface areas used in Table 4-13 are derived from Poppendorf and Leffingwell 1982. Additional values for other parts of the body may be found in Appendix H. These surface areas may be used in some special circumstances.

In Table 4-13, the amount per cm^2 of skin contacted is derived from Versar 1984. In that study, immersion of the hands in liquids of varying viscosity was found to result in retention of 5 to 14 mg liquid per cm^2 surface area (increasing with increasing viscosity). This range is appropriate for activities where there is frequent, required contact with the material. Versar also reported results generally ranging from 1 to 3 mg/cm^2 for activities such as wiping up a spill or wiping the hands with a contaminated cloth, and for the amount remaining when the hands were "partially" wiped with a clean cloth after immersion. This range is appropriate for situations where there may be contact, but immersion is not expected.

Versar evaluated immersion by immersing a hand in a jar of liquid, allowing excess liquid to drip into the jar, and weighing the jar to determine how much liquid was removed. Six liquids of various viscosities were used. The data for immersion of the hand range from 4.99 to 13.85 mg/cm^2 (for water and mineral oil, respectively). The concentration was a direct function of the kinematic viscosity of the liquid being tested. After partial wiping, the range was 1.3 to 2.72 mg/cm^2, and was not related to viscosity. The lowest quantities remaining after partial wipe were for the more viscous liquids tested.

The data for handling a rag ranged from 1.37 to 3.88 mg/cm^2. The least viscous materials had the highest data points. Data for cleaning a spill ranged from 0.67 to 1.11 mg/cm^2 and were not related to viscosity. Partial wiping of the hand afterward resulted in the remaining liquid ranging from 0.31 to 3.44 and 0.48 to 1.27 g/cm^2, respectively. Full wiping after the handling of a rag reduced the amounts remaining to 0.01 to 3.3 mg/cm^2.

The tests for initial and secondary uptake gave results of 1.2 to 3.0 mg/cm^2 for the initial quantity on the hand. Again, the highest values were for the least viscous materials. Partial and full wipes reduced the amounts to 0.1 to 2.7 mg/cm^2.

These data suggest that for contact with rags and for liquid remaining after partial wiping when the hands have been immersed, the amount of material on hands ranges from 1 to 3 mg/cm^2 in nearly all cases, as concluded by Versar. No relationship to viscosity is obvious. For immersion (without wiping) values ranged from about 5 to 14 mg/cm^2.

The Versar data are the most complete known for liquids. A PMN submitter measured immersion data and found a similar relationship to viscosity when gloved hands were used.

Kin Wong evaluated wiping both hands with glycerin, paraffin, oil, and water. He obtained values of 0.5 to 1.8 g on the hand, equivalent to 0.4 to 1.4 mg/cm^2, assuming 1300 cm^2 surface area for two hands.

3. <u>Duration of Contact</u>

The inclusion of duration of exposure must acknowledge, if not quantify, several mechanisms affecting dermal absorption. First is the issue of volatilization of some (if not most) of the initial deposit from the skin before complete absorption can take place. Second are workplace factors affecting the dermal absorption rate. These include humidity (skin hydration), dermatitis, abrasion, and clothing practices. Some consideration also should be given to the presence or absence of chemical warning properties such as visual changes to the skin, irritation, or corrosive properties. The engineer must check that an estimated dermal exposure does not exceed the amount of material that could be present on the worker's skin.

The amount of time for which the contact can occur should generally be estimated as 1 to 4 hours, based on the expectation that the worker will, at a minimum, thoroughly wash the hands at lunch or end of the day. However, volatile materials may evaporate rapidly from the hand. The actual duration of dermal contact therefore may not exceed the duration of the activity leading to the contact. For example, the volatilization model used to estimate airborne concentrations of materials can be used to show that a thin layer of toluene will evaporate in several minutes. If an activity such as sampling were estimated to require half an hour, this duration can be used as the maximum exposure time. The number of contacts per day can be estimated as one or more, depending on whether the worker is expected to handle the chemical throughout the day or the chemical is rapidly absorbed but replenished by additional contacts. This is based on expecting the worker to wash up at meal breaks. When two distinct periods of contact are expected, the duration, but not the quantity, of potential contact should be doubled. That is, if a worker's activities involving a particular chemical last throughout the day, the engineer should report contact as potentially lasting 8 hours, but totalling only the estimate for a single contact.

This method provides an estimate of the amount of material on the skin. The Health and Environmental Review Division (HERD) usually estimates absorption of the material through the skin as a percent of the amount of contact. Sometimes the flux of a material through the skin may be known or estimated in terms of mg absorbed per cm^2 per unit of time. In such cases, the amount absorbed could be calculated based on CEB's estimate of area contacted and duration of contact, not the amount on the skin.

C. Personal Protective Equipment

The primary types of personal protective equipment used to reduce worker exposure are gloves and respirators. Aprons, coveralls, goggles, and face shields also may be used where necessary.

1. Gloves

The ability to predict glove use practices, especially for end-users of a chemical, is poorly established. Glove protection depends both on glove selection and work practices. Therefore, it is difficult to define the level of protection resulting from the general use of gloves. There are no gloves available that are totally impervious to chemicals. Also, there are no existing models to predict the degree of protection offered by a glove of a particular material when used in contact with a specific chemical. However, if gloves are properly selected and appropriately used (considering available information on permeation, penetration, degradation, and frequency of replacement), their use will significantly reduce exposure. The effectiveness of gloves is a function of characteristics of the glove, the type of chemical to which exposure may occur, the conditions of exposure, and the activities of the worker.

The permeation rate is the rate at which a chemical moves through a glove material. It is normally measured by testing a small piece of glove material using a standard method such as ASTM F-739-81, Resistance of Protective Clothing Materials to Permeation by Hazardous Liquid Chemicals. Breakthrough times for many chemicals are listed in A.D. Little 1985. Where many breakthrough times are consistent for a chemical or chemicals, this may be referred to as a guide for breakthrough times for that chemical.

Glove materials (plastics or rubber) are not impermeable. Permeation tests of intact glove material usually result in the determination of a breakthrough time (i.e., the time between start of the test and the first detection of the test chemical in the test chamber). Only after breakthrough has occurred can steady-state permeation rates be established. The ASTM method aims to establish this steady-state rate.

Gloves or glove materials are also subject to penetration (ASTM definition: "flow-through zippers, seams, pinholes, or other imperfections") and to degradation by chemical or mechanical means. Permeation rates do not refer to these phenomena, although a material undergoing a permeation test may degrade to the point that the test is invalid.

Requirements for gloves also must address the conditions of exposure (e.g., one hour with the glove immersed in the chemical). A sufficiently long breakthrough time relative to the duration of exposure may mean that a glove prevents exposure as long as it is intact and undegraded. The glove may be determined to be "impervious" because exposure is too short for the chemical to achieve breakthrough.

The ASTM method states that resistance is determined by measuring the breakthrough time and monitoring the subsequent permeation rate. A "resistant" material is not defined in the standard. Glove makers do rate the resistance of gloves subjectively as "excellent, good," etc. Unless "resistance" is very strictly defined, this is a less stringent requirement than that the material be "impervious." A material that resists breakthrough for 30 minutes may be rated resistant though exposure may occur for hours.

"Guidelines for the Selection of Chemical Protective Clothing" contains a matrix of glove materials rated for many chemicals (A.D. Little 1985). The guideline provides qualitative recommendations for twelve common glove materials based on both computer predictions and test data. The recommendations are presented by chemical and chemical class. This matrix should be used only as guidance.

A computer model that predicts the amount of a chemical that will permeate glove materials has been developed by A.D. Little (A. D. Little 1989). It is designed to predict the amount of permeation through five different glove materials for some organic chemicals. The glove materials are natural rubber, nitrile rubber, neoprene, low density polyethylene, and butyl rubber. Material thickness can be varied. Besides gloves, the model can predict permeation rates for aprons, coveralls, and other protective clothing that are constructed of one of the five materials.

The model requires chemical-specific input: molecular weight, density, and vapor pressure or molecular structure of the chemical. The model also incorporates specialized versions for certain chemical groups for which it requires more detailed input resulting in a more accurate prediction. The output is a cumulative amount of chemical permeating through the material over time, presented as mass per surface area for a designated period. The model also provides breakthrough times.

The following language has been adopted as standard for requiring the use of gloves in orders written under Section 5 of TSCA:

Workers who may be dermally exposed to the PMN substance shall wear gloves determined by the Company to be impervious to the PMN substance under the conditions of exposure, including the duration of exposure. The Company may decide this either by testing the gloves under the conditions of exposure or by evaluating the specifications provided by manufacturers of the gloves. Testing or evaluation of specifications should include consideration of permeability, penetration, and potential chemical and mechanical degradation by the PMN or associated materials.

A requirement for gloves usually should not specify their length (e.g. elbow length). For most exposure scenarios, it seems reasonable to require protection of the hands when there is potential that they would be immersed in a PMN chemical or otherwise contact bulk amounts of the chemical. Longer gloves should only be needed where there is considerable potential for splashing of the material. In such cases, aprons or other body covering also may be in order. The engineer on the case may be in the best position to make recommendations in this area, based on familiarity with the processes that may lead to extensive exposure. It should be noted that CEB does not consider the physical environment in which the gloves will be worn, but considers only the potential for chemical permeation, penetration, and degradation. The physical environment is an important consideration as gloves are often composed of polymeric materials that may be stressed, punctured, or otherwise damaged in actual used.

2. <u>Respirators</u>

Respiratory protection for new and existing chemicals presents some unique and complicated problems for the assessment and control of occupational exposure. Often, the potential for inhalation exposure is a primary concern for the workers who handle new or existing chemicals.

a. <u>OSHA Requirements</u>

OSHA regulations in 29 CFR 1910.134 cover the proper use, selection, care, and maintenance of respiratory protection. OSHA Permissible Practice requires that engineering controls are first considered for reducing worker exposure levels. However, if engineering controls are not feasible, need to be supplemented, or are in the process of being instituted, appropriate respiratory protection should be used to reduce worker exposure. Use of even one respirator by one employee requires the implementation of a respirator program that meets the minimum program elements specified by OSHA. Besides the use of an appropriate NIOSH or Mine Safety and Health Administration (MSHA) approved respirator, the minimum requirements for this respirator program are summarized as follows:

- Written standard operating procedures.
- Selection based on the hazards to which the worker is exposed.
- Training and instruction.
- Provisions for cleaning, disinfection, storage, and maintenance.
- Respiratory Program inspection and evaluation.
- Determination that employees are physically able to perform the work and use the respiratory protective equipment.

Further respirator use, selection, and program details are available in the American National Standards Institute (ANSI) Standard Z88.2 1990 (in publication). Details on respirator construction and performance testing are covered in the Bureau of Mines regulations at 30 CFR 11 (latest revision 7/28/89). It should be noted that the OSHA standard (29 CFR 1910.134) and the NIOSH certification standards (30 CFR 11) are both being revised at this time.

b. Selection of a Respirator

(1) Background

Industrial hygienists go through a series of decisions to select the appropriate respirator. These include:

- The desired exposure level (e.g., TLV);
- The ambient concentration of the chemical in the work environment; and
- The specific work conditions under which the respirator will be worn.

These and other human factors are discussed more fully in Appendix I.

NIOSH has developed a document entitled "A Guide to Industrial Respiratory Protection" which contains a decision logic for assessing the respiratory protection requirements under many conditions (NIOSH 1976). This decision logic was updated in 1987 (NIOSH 1987). This reference should be consulted when selecting respirator needs.

TLVs are rarely established for new chemical substances and no information about absorption of the chemical by carbon in organic vapor cartridges or odor threshold is typically available. Selection of an appropriate respirator is difficult without this information. Therefore, the rationale of selection should be clearly presented. It also must be remembered that wearing a respirator is often a burden to a worker performing tasks. Not all worker activities or types of respirators are amenable to use of respirators over extended periods of time.

(2) <u>Selection of Respiratory Protection by CEB</u>

Selection of respiratory protective equipment by CEB is constrained by many factors as described in Appendix I. For CEB, selection is limited to recommending different classes or categories of respirators based on state-of-the-art knowledge of respiratory protection. The document "Strategy for Recommending Respirators for Control of Exposure to Substances Undergoing Premanufacturing Notice (PMN) Review" (Myers nd), which is contained in Volume II, should be used to select the appropriate class of respiratory protection for a PMN chemical. The selection of a particular device within a recommended category must be made by a knowledgeable person at the workplace who is familiar with the actual conditions of use. When respiratory protection is required, industrial hygienists typically weigh many factors into the consideration of alternative respirator devices including:

- The permissible exposure limit (PEL) specified by OSHA or threshold limit value (TLV) specified by the American Conference of Governmental Industrial Hygienists (ACGIH) or other allowable exposure level established.
- The warning properties of the contaminant.

- The concentration of the contaminant (and other contaminants) in the work area.
- The type of hazard (gas, fume, mist, particulate, etc.).
- The degree of protection afforded by various respirators.
- The conditions of use of the respirator including other hazards present, such as lack of oxygen or confined space.
- The workers' ability to wear a particular device.
- Worker comfort, degree of wear time, need for communication or specific duties that may be affected by the respiratory device.

For PMN substances, there is generally no established PEL or TLV and it is generally unknown whether there are adequate warning properties. According to 30 CFR 11, air-purifying respirators are prohibited for protection against organic vapors with poor warning properties unless there is an OSHA or other Federal standard that permits their use. OSHA has allowed the use of air-purifying devices for substances with poor warning properties (e.g., acrylonitrile, benzene, and vinyl chloride), if cartridges or canisters are changed prior to the end of the service life (before breakthrough) or before the beginning of the next shift, whichever comes first (29 CFR 1910.1017, 1910.1028, and 1910.1045).

The selection of an air-purifying, organic vapor respirator requires a complex analysis of the anticipated performance of the chemical cartridge or canister in atmospheres containing the substance of concern. CEB must consider the following factors in this analysis:

- Human levels of detection (odor threshold) or properties that make recognition easy (irritation, lacrimation).
- How long it takes the PMN substance, at the concentration in the workplace, to break through the sorbent (cartridge service life).
- The expected duration of exposure compared to the cartridge service life.
- How often and when cartridges will be replaced.
- Reaction products or amount of heat generated during sorption.

The consideration of an organic vapor respirator for a PMN substance should be based on the following:

- A decision by Division Directors that a protection factor of 50 or less will afford adequate protection for the PMN substance.
- An initial screening by CEB that determines that the PMN is a good candidate for an air-purifying organic gas and vapor respirator based on consideration of:
  - Warning properties.
  - Possibility for heat generation or generation of toxic chemicals upon sorption.
  - Results from models that predict cartridge service life.
  - Other sources of information, such as ACGIH, NIOSH, American Industrial Hygiene Association (AIHA recommendations for analogues, etc.

If CEB determines that a PMN substance is not a good candidate, the 5(e) Order would specify a supplied-air respirator, but the company could petition EPA and provide the necessary documentation proving that an air-purifying respirator is acceptable and that the organic vapor cartridge provides acceptable performance.

In either case, the submitter would be required to document the effectiveness of an organic vapor cartridge respirator by submitting the following information:

- The service life of the cartridge.
- Information showing that high temperatures (>40°C) or toxic chemicals are not generated upon sorption of the PMN substance.
- A respirator cartridge change-out schedule established as part of the respirator program.
- Any known information on the warning properties of the PMN substance.
- Any administrative controls that will be used by the submitter.

NIOSH is preparing guidance to be used for testing organic gas/vapor cartridges for service life. However, until this guidance is available, CEB recommends that organic gas/vapor cartridges be tested in accordance with the proposed NIOSH 42 CFR 84, Tests and Requirements for Certification of Permissibility of Respiratory Protective Devices Used in Mines and Mining, published in the Federal Register on August 27, 1987. A CEB industrial hygienist should be consulted to determine the appropriate test concentrations for the PMN substance, and answer any questions the submitter may have.

This information should be submitted for CEB review before OTS approves use of an air-purifying respirator. This occurs either before an order is written or upon a company's request to modify an existing order. Submitted information will be reviewed by a CEB industrial hygienist on a case-by-case basis.

(3) Protection Factors

Protection factors for various classifications of respirators have been assigned by NIOSH (see Appendix J). CEB uses these values will be used to decide the appropriate type(s) of devices necessary to achieve the degree of protection determined by Division Directors.

The protection factor of a respirator is an expression of the performance based on the ratio of the concentration of contaminant measured outside the facepiece cavity to the concentration of contaminant measured inside the facepiece cavity. The Assigned Protection Factor (APF) is defined by NIOSH as a measure of the minimum anticipated workplace level of respiratory protection that would be provided by a properly functioning respirator or class of respirators to a percentage of properly-fitted and trained users.

The table of protection factors developed by NIOSH may be simplified since most classes of respirators have Assigned Protection Factors that fall into one of four categories: APF > = 2000; APF > = 50; APF > = 25; and APF > = 10. Thus, respirators will be selected based on the need for a 1000-, 50-, or 10-fold reduction in the estimated potential inhalation exposure.

<u>APF of at Least 2000 is Considered</u>. Devices which have been assigned a protection factor of 2000 or greater are supplied-air respirators equipped with a full facepiece and operated in positive pressure mode. Since the selection of a supplied-air respirator is independent of the physical and chemical properties of the contaminant, no further information is required to make this choice.

<u>APF of at Least 50 is Considered</u>. Supplied-air devices that have been assigned a protection factor of at least 50 are equipped with a tight-fitting facepiece and operated in continuous flow mode, or are equipped with a full facepiece and operated in demand (negative pressure) mode. However, OSHA does not allow the use of supplied-air respirators operated in demand mode. Supplied-air respirators operated in positive pressure mode have much better performance as facepiece leakage is minimal with positive pressure. Thus, EPA will allow the use of positive pressure, supplied-air respirators only. Air-purifying respirators equipped with a full facepiece and powdered air-purifying respirators equipped with a tight fitting facepiece have been assigned protection factors of 50.

<u>APF of at Least 25 is Considered</u>. Powered air-purifying respirators equipped with a loose fitting hood or helmet have been assigned a protection factor of 25.

<u>APF of at Least 10 is Considered</u>. Most approved respirators have a protection factor of at least 10. For a particulate exposure, a high efficiency particulate filter is required. To recommend the use of an organic vapor respirator, the effectiveness of the cartridge or canister must be evaluated as described above.

(4) Standard Language for 5(e) Orders

The standard language as presented in Appendix J reflects the use of the NIOSH APF values. It describes the general classes of respiratory protection that should be recommended in 5(e) or other orders. Any questions regarding respiratory protection for 5(e) Orders should be referred to a CEB industrial hygienist.

D. Engineering Controls

1. Local Exhaust Ventilation

Local exhaust ventilation (LEV), in addition to general ventilation, is the primary control used to reduce worker exposure to chemicals. If LEV is used for a specific activity, the specifications of the control should be compared to the recommendations made by ACGIH in "Industrial Ventilation: A Manual of Recommended Practice" (ACGIH 1986). Although a qualitative determination of control use can be made, it will usually not be possible to account quantitatively for local exhaust.

The actual reduction in worker exposure from the use of LEV is not a simple relationship. It is dependent on several factors:

- The design capture efficiency of the LEV system;
- The work practices of the employee;
- Air currents in the work area;
- Other process-specific factors such as dragout or sudden releases; and
- Actual maintenance of the LEV system over time.

The design capture efficiency is primarily dependent on hood design and exhaust volume. The best hood design is one that encloses or confines the process. This is not always possible when worker access to the process is considered. Exhaust volumes (or face velocities) presented in the ACGIH ventilation manual for a similar process should be compared with information supplied by the submitter to ensure that adequate face velocity will be used in the design.

The addition of an LEV system can affect the work practices of the employee. Enclosure of the process can cause changes in work practices that may make the LEV less effective. The LEV design should never allow the breathing zone of the worker to pass between the pollutant source and the exhaust.

Air currents in the work place from other processes, natural ventilation, general ventilation, or movement in the area can reduce the effectiveness of an LEV system. All possible cross currents should be minimized.

Process-specific factors such as dragout can cause the chemical to be carried out of the capture zone of the LEV system, thus decreasing its efficiency. Processes that release sudden surges of hot gases or vapors must design the LEV system to account for these releases.

Finally most LEV systems are not maintained at peak efficiency throughout their life. If a system is older, the questions of design parameters such as face velocity versus actual parameters should be addressed.

One process where standard LEV design can obtain very high efficiencies is the use of lay-on or slot LEV to control worker exposure during drum filling. In a series of tests on LEV use in drum filling performed for OTS, capture efficiencies were between 99 to 100 percent when the ACGIH design flow rate was used (PEI 1987). The efficiency of the lay-on and slot LEV systems in drum filling operations generally was independent of the concentration of emissions leaving the drum. The efficiency was affected by the fill rate that determines the emission velocity at the drum bung hole.

2. Nitrogen Blanketing

Chemicals that are extremely volatile or that react with water or air on contact are often loaded into shipping containers under a nitrogen blanket. This handling procedure (known as nitrogen blanketing, padding, or purging) displaces air and moisture and therefore prevents degradation of the chemical and decreases the chance of explosion. Volume II contains a description of the process and work activities associated with the procedure for acetaldehyde, hydrazine, and toluene diisocyanate (Mitre 1984).

The only release expected from nitrogen blanket transfer is a small spill from the end of the transfer line of the liquid remaining between the valve and the end of the coupling collar. Assumptions to be made in the absence of other information are described in the report and include:

- Nominal inside diameter of transfer line equals 3 inches.
- Distance between valve body and end of collar equals 3 inches.
- Area of spill is assumed to be 730 cm^2.

In the original document describing nitrogen blanketing (Mitre 1984), a model was assumed for the generation rate and an empirical equation for the mass transfer coefficient. In this revision of the engineering manual, CEB has revised the model for generation rate from an open surface. Using the equation for estimating airborne concentration from an open surface (Equation 4-25) and substituting the surface area of the spill:

Worst case:

$$C_v = 1679\ P \qquad \text{Equation 4-27}$$

Typical case:

$$C_v = 54.8\ P \qquad \text{Equation 4-28}$$

where: C_v = Airborne concentration, ppm
P = Vapor pressure, mm Hg

3. Transfers

Emissions from controlled loading operations can be calculated by multiplying the uncontrolled emission rate (Equation 4-21) by the control efficiency term:

$$G = \left(1 - \frac{E}{100}\right) \frac{f\ M\ V\ r\ P^{\circ}}{3600\ R\ T_L} \qquad \text{Equation 4-29}$$

where G = Vapor generation rate, g/sec

E = Control efficiency, percent

f = Saturation factor, dimensionless

M = Molecular weight, g/g-mol

V = Volume of container, cm^3

r = Fill rate, units/hr

P° = Vapor pressure of pure substance, atm

R = Universal gas constant, 82.05 atm cm^3/g-mol·K

T_L = Liquid temperature, °K

Measures to reduce loading emissions include use of vapor recovery equipment to capture the vapors displaced during loading, recovery by refrigeration, absorption, adsorption or compression, and piping back to storage. Vapors also can be controlled through combustion in a thermal oxidation unit with no product recovery. Control efficiencies of modern units range from 90 percent to over 99 percent depending on the nature of the vapors and the type of controls used (USEPA 1985b).